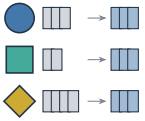
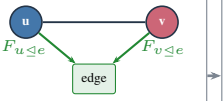
1 Encode nodes



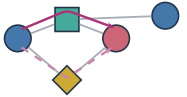
per-type encoder $\longrightarrow h_v$
+ context c_{t_v}
(mean per relation)

2 Restriction maps



learned from both endpoints;
weight a_e per edge
neighbours are compared after the maps (heterophily)

3 Sample faces



random walks on the original graph;
one walk = one face

no meta-path list;
unbiased weight m_f

4 Face maps & weight



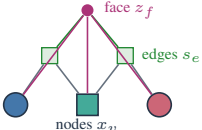
face maps = edge maps composed along the path

forced: no new parameters

$\beta_f = m_f \exp g(\xi_f)$

from the face's own nodes and relation sequence

5 One energy



node–edge, edge–face, node–face:
every pair compared after its restriction maps

6 Propagate & predict

$Y^\star = \arg\min_Y \ \frac{1}{2}\|x - h\|^2 + \frac{\tau}{2}\, Y^\top \hat{L}\, Y$

one energy, unique minimiser for every parameter value

$Y = (x, s, z)$: node, edge and face states

$\hat{y}_v = \mathrm{MLP}([h_v, x_v^\star])$

encoding and propagated state